

INTERNPATh : A Smart Internship Guidance & Readiness System

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Abstract—Finding suitable internships remains a significant challenge for students due to the lack of personalized guidance, inadequate awareness of industry requirements, and the limitations of existing internship platforms, which primarily provide generic listings without evaluating a student's suitability for specific opportunities. This often results in inefficient application processes and reduced chances of securing relevant internships. This paper presents **InternPath**, an intelligent internship guidance and readiness platform designed to streamline internship discovery and improve student preparedness for career opportunities. Built using a modern web architecture comprising React.js, FastAPI, and PostgreSQL, the system aggregates internship opportunities from multiple online sources through automated web scraping and maintains a centralized repository for efficient data management and analysis. The platform incorporates a skill-based recommendation engine that matches student profiles with internship requirements to generate personalized and ranked internship suggestions. To enhance application quality, Natural Language Processing (NLP) techniques utilizing spaCy and NLTK are employed for resume analysis, extracting key information such as skills, education, and project experience. Furthermore, a rule-based Internship Readiness Scoring mechanism evaluates student preparedness and provides actionable recommendations for improvement, while a fake internship detection module identifies potentially fraudulent or suspicious opportunities. The system also integrates an AI-powered chatbot to provide instant guidance and support throughout the internship search process. Experimental deployment demonstrates that InternPath significantly improves internship discovery efficiency, enhances resume quality, and supports informed career decision-making through a unified and intelligent platform.

Index Terms—Internship Recommendation System, Natural Language Processing, Resume Analysis, Internship Readiness Scoring, Web Scraping, Fake Internship Detection, React.js, FastAPI, PostgreSQL.

I. INTRODUCTION

Internships play an important role in the career development of students by providing practical exposure to real-world

industry environments. Through internships, students can apply the theoretical knowledge learned during their academic studies to practical tasks and gain valuable professional experience. However, many students face difficulties in identifying suitable internship opportunities that match their skills, interests, and academic qualifications. The process of searching for internships is often time-consuming because students need to explore multiple platforms and analyze the requirements of each opportunity manually. Most existing internship platforms mainly provide lists of available internships but do not offer personalized recommendations or guidance. As a result, students may apply for internships without knowing whether they meet the required qualifications. In addition, the presence of fraudulent internship opportunities online makes it difficult for students to determine which opportunities are trustworthy. Therefore, there is a need for a system that can guide students in finding relevant internships and help them evaluate their readiness before applying. The InternPath system is developed to address these challenges by providing a smart internship guidance platform. The system integrates technologies such as web scraping, Natural Language Processing, and artificial intelligence to recommend internships based on student skills, analyze resumes, evaluate internship readiness, and detect potentially fake internship opportunities.

II. SYSTEM ARCHITECTURE

The InternPath system is designed as a web-based platform that assists students in discovering internships, analyzing their readiness, and improving their profiles. The system architecture consists of several interconnected components that work together to collect internship data, process user information, and generate intelligent recommendations. The system integrates web scraping, database management, Natural Language Processing, and artificial intelligence to provide

a comprehensive internship guidance solution. As shown in Fig. 1, the student can register or log in, update their profile, view personalized internship recommendations, check their internship readiness score, interact with the chatbot for guidance, and verify the authenticity of internships. This diagram highlights the key services provided by the system to support students in their internship search and preparation process.

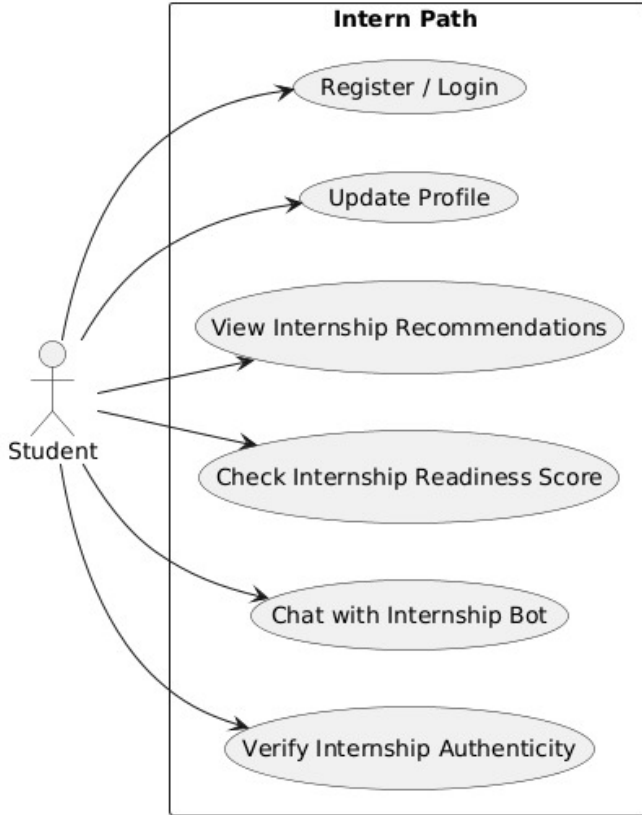


Fig. 1. Internpath System Core Use Case Specification.

A. Multi-Tier Architectural Breakdown

The InternPath system is designed as a web-based platform that assists students in discovering internships, analyzing their readiness, and improving their profiles. The system architecture consists of several interconnected components that work together to collect internship data, process user information, and generate intelligent recommendations. The system integrates web scraping, database management, Natural Language Processing, and artificial intelligence to provide a comprehensive internship guidance solution:

- **Data Collection via Web Scraping:** The system collects internship data from Internshala website using web scraping techniques. Tools such as browser automation and scraping libraries are used to extract relevant information including internship role, company name, required skills, location, stipend, and application details. The collected data is cleaned and structured before being stored in the database.

- **Database Management:** A centralized database is used to store the scraped internship data and user-related information. The database maintains records of internship listings, student profiles, resume analysis results, and readiness scores
- **Resume Analysis Module:** The resume analysis module allows students to upload their resumes in PDF or DOC format. Natural Language Processing techniques are used to extract important information such as skills, education, and project experience. The system analyzes the resume structure and keywords to generate a resume quality score and provides suggestions for improvement.
- **Internship Recommendation Engine:** The recommendation engine matches student skills with internship requirements stored in the database. The system calculates a skill match percentage and ranks internships according to relevance. Students can filter recommendations based on preferences such as domain, location, or stipend.
- **Internship Readiness Scoring:** The system evaluates a student's preparedness for internships using a rule-based scoring mechanism. Factors such as academic performance, skills, projects, and experience are analyzed to generate a readiness score. The score is displayed along with a readiness level such as Low, Medium, or High.
- **Fake Internship Detection:** The fake internship detection module helps identify potentially fraudulent internship postings. When a user submits an internship URL, the system checks domain credibility, suspicious keywords, and missing contact information. Based on these checks, the system assigns a risk level indicating whether the internship appears trustworthy or suspicious.
- **AI Chatbot Assistance:** The system includes an AI-powered chatbot that assists students with internship-related queries. The chatbot retrieves relevant information from the internship database and provides context-aware responses. This feature helps students receive guidance about internships, skills required for specific roles, and career preparation.
- **User Interface:** The frontend interface provides an interactive and user-friendly environment where students can access all system features. The interface includes pages such as the student dashboard, internship listings, resume analysis, readiness score display, fake internship detection, and chatbot interaction.

III. METHODOLOGY AND TECHNICAL IMPLEMENTATION

The proposed InternPath system consists of three major modules: Data Collection via Web Scraping, NLP-Based Resume Analysis, and Rule-Based Fake Internship Detection. These modules work together to provide personalized internship recommendations, evaluate student readiness, and identify potentially fraudulent internship opportunities.

A. Data Collection via Web Scraping

The internship recommendation process begins with collecting internship data from online internship platforms. Web scraping techniques are employed to automatically extract internship information such as internship title, company name, required skills, stipend, duration, location, and application links. Python libraries including BeautifulSoup, Requests, and Selenium are used for data extraction and dynamic content handling.

The extracted data is cleaned to remove duplicate and incomplete entries before being stored in the internship database. Periodic scraping ensures that the dataset remains updated with newly posted internships.

Let the set of internships be represented as:

$$I = \{i_1, i_2, i_3, \dots, i_n\} \quad (1)$$

where each internship record is defined as:

$$i_k = \{T_k, C_k, S_k, L_k, D_k, St_k\} \quad (2)$$

where:

- T_k = Internship Title
- C_k = Company Name
- S_k = Required Skills
- L_k = Location
- D_k = Duration
- St_k = Stipend

The collected dataset serves as the primary knowledge source for the recommendation system.

B. NLP-Based Resume Analysis

To understand a student's profile and capabilities, the uploaded resume is processed using Natural Language Processing (NLP) techniques. The resume text is extracted from PDF or DOCX files and undergoes several preprocessing stages, including tokenization, stop-word removal, punctuation removal, and lemmatization.

After preprocessing, important information such as technical skills, programming languages, certifications, and educational details are identified. These extracted skills are compared with internship requirements to determine compatibility.

Let the set of skills extracted from a resume be represented as:

$$R_s = \{r_1, r_2, r_3, \dots, r_m\} \quad (3)$$

Similarly, let the required internship skills be:

$$I_s = \{s_1, s_2, s_3, \dots, s_n\} \quad (4)$$

The skill matching score is computed as:

$$\text{Skill Match Score} = \frac{|R_s \cap I_s|}{|I_s|} \times 100 \quad (5)$$

where $|R_s \cap I_s|$ represents the number of common skills between the resume and internship requirements.

The Internship Readiness Score is then calculated by combining multiple factors:

$$IRS = w_1M + w_2E + w_3P \quad (6)$$

where:

- M = Skill Match Score
- E = Experience Score
- P = Profile Completeness Score
- w_1, w_2, w_3 = Weight factors

This score helps students assess their preparedness before applying for internships.

C. Rule-Based Fake Internship Detection

To ensure user safety, the system includes a fake internship detection module. The module evaluates internship listings using predefined rules based on common characteristics of fraudulent postings.

Some of the evaluated indicators include:

- Missing company information
- Suspicious URLs
- Unrealistically high stipends
- Incomplete job descriptions
- Lack of contact information
- Excessive personal information requests

Each rule contributes to a risk score.

Let

$$R_i = \begin{cases} 1, & \text{if rule } i \text{ is violated} \\ 0, & \text{otherwise} \end{cases} \quad (7)$$

The overall fake internship score is computed as:

$$FS = \sum_{i=1}^n w_i R_i \quad (8)$$

where:

- FS = Fake Internship Score
- w_i = Weight assigned to rule i
- R_i = Rule outcome

An internship is classified as suspicious if:

$$FS > \theta \quad (9)$$

where θ is a predefined threshold.

If the threshold is exceeded, the internship is flagged and a warning is displayed to the student. This transparent rule-based approach enables easy interpretation and modification of detection criteria.

D. Personalized Internship Recommendation

Based on the internship database and resume analysis results, the system generates personalized internship recommendations. The recommendation engine ranks internships according to skill compatibility and readiness score.

The recommendation score is computed as:

$$RS = \alpha M + \beta IRS \quad (10)$$

where:

- RS = Recommendation Score
- M = Skill Match Score
- IRS = Internship Readiness Score
- α, β = Weight parameters

Internships with higher recommendation scores are prioritized and presented to the student.

E. AI Chatbot Assistance

An AI-powered chatbot is integrated into the system to assist students with internship-related queries. The chatbot utilizes Retrieval-Augmented Generation (RAG) techniques to retrieve relevant internship information and provide contextual responses. Students can seek guidance regarding resume preparation, internship eligibility, skill requirements, application procedures, and interview preparation. The chatbot enhances user engagement and provides immediate assistance without requiring human intervention.

IV. SUSTAINABLE DEVELOPMENT AND SOCIAL RELEVANCE

The proposed InternPath system contributes to sustainable development by promoting quality education, employability, and equal access to career opportunities for students. Many students, particularly those from rural and underserved backgrounds, face difficulties in identifying suitable internships due to limited guidance, lack of awareness, and insufficient industry exposure. By providing personalized internship recommendations based on skills, academic background, and readiness level, the system helps bridge the gap between education and employment.

The Internship Readiness Score encourages continuous skill development and self-assessment, enabling students to identify and improve areas where they may be lacking. The AI-powered chatbot provides accessible career guidance at any time, reducing dependence on manual counseling and making mentorship support available to a larger number of students. Furthermore, the fake internship detection module protects students from fraudulent opportunities, creating a safer and more trustworthy internship application environment.

The project aligns with the United Nations Sustainable Development Goals (SDGs), particularly SDG 4 (Quality Education) by supporting skill development and career readiness, SDG 8 (Decent Work and Economic Growth) by improving access to internship opportunities and enhancing employability, and SDG 10 (Reduced Inequalities) by ensuring that students from diverse educational and socioeconomic backgrounds receive equal access to internship guidance and career resources. By helping students make informed career decisions and improving their chances of securing relevant internships, InternPath contributes to the development of a skilled workforce and supports long-term economic and social growth.

V. CONCLUSION AND FUTURE SCOPE

The project successfully developed a smart internship guidance system that assists students in finding suitable internship opportunities efficiently and reliably by collecting real-time internship data through web scraping, ensuring that recommendations remain relevant and up to date. The system provides personalized internship suggestions based on students' skills, semester, and readiness level, while the Internship Readiness Score helps students assess their preparedness before applying, thereby reducing unnecessary rejections. An AI-powered chatbot supports users by answering internship-related queries, and a fake internship detection feature improves safety by identifying potentially fraudulent opportunities. The use of a rule-based approach ensures transparency, simplicity, and ease of implementation. Overall, the project achieved its objective of simplifying the internship search process, enhancing decision-making, and supporting students in better career planning. In the future, the system can be expanded by integrating multiplatform scraping from sources such as LinkedIn, Naukri, and Unstop to provide a broader range of internship opportunities. Recommendation capabilities can be improved through machine learning techniques such as collaborative filtering and content-based filtering, while fake internship detection can be enhanced using classification models like Support Vector Machines and Random Forests trained on labeled datasets. Additional features such as an AI-powered ATS-friendly resume builder, real-time internship alerts, an interview preparation module with mock tests and domain-specific guidance, an application tracking system, and a college analytics dashboard can further improve the platform's functionality, making it a comprehensive career development and internship management solution for students and educational institutions.

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